GUILHERME CHAGAS MUNDT

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Alegrete, Rio Grande do Sul, Brazil

Summary

Software Engineering undergraduate at the Federal University of Pampa, currently building software for a public health unit as a funded fellow (React.js, NestJS, Docker), after deploying a management platform in a real school as my curricular internship. Experience spans full-stack web development (React, Node.js, NestJS, SvelteKit) and applied AI systems (RAG pipelines, LLM agents and explainable machine learning), always in teams working under Git Flow with peer-reviewed code. Looking for a junior software engineer or web developer position where I can contribute to production software and keep growing as an engineer.

Experience

- **PET-Saúde Group** *09/2025 - Present*
  Software Developer (Funded Fellow, Part-time) Hybrid, Brazil
  - Build technology solutions for the public health unit of Jaguari (RS) as a funded fellow, with React.js on the frontend and NestJS on the backend.
  - Delivered a citizen communication system covering prenatal and childhood vaccination, keeping residents informed about their immunization schedule.
  - Currently developing the patient transport and medical exam modules of the health unit's system; containerize services with Docker and support their deployment.
- **BRISA — Society for the Development of Information Technology** *03/2025 - 09/2025*
  Software Development Resident (Residência em TIC, Part-time) Hybrid, Brazil
  - Completed a seven-month industry residency, working in a team on a real system under professional development practices.
  - Built REST services in TypeScript (Node.js/Express) with a controller/route/service layering and Git Flow branching with peer-reviewed pull requests.
- **Demétrio Ribeiro School — Mandatory Curricular Internship [ ]** *03/2026 - 12/2026*
  Software Developer Alegrete, Brazil
  - Deployed the Scavenger Project school competition platform at the school, taking a system built in coursework into real use by staff and students.
  - Developed the full stack — React frontend and Node.js/Express backend over MongoDB Atlas — covering teams, events, penalties, feedback and notifications, with JWT-protected routes and transactional e-mail delivery.
  - Largest contributor among eight developers (74 commits and 24 of the 28 pull requests), working under Git Flow with peer-reviewed pull requests.

Education

- **Federal University of Pampa (UNIPAMPA)** *03/2023 - Present*
  Bachelor of Software Engineering Alegrete, Brazil
- **Carlos Drummond de Andrade** *12/2022*
  Secondary Education Alvorada, Brazil

Projects

- **UniBot — Institutional Conversational Assistant (RAG)** *05/2026 - 06/2026*
  Tools: Python, FastAPI, Agno, ChromaDB, Groq, Tavily, React, Vite, TypeScript, Pytest [ ]
  - Four-developer course project: an assistant that answers questions about UNIPAMPA institutional documents (course plan, resolutions, academic calendar) through RAG with hybrid retrieval over ChromaDB.
  - Built the external web-search agent (Agno, Groq, Tavily) with a pluggable search provider, structured output with citations and retries for malformed tool calls.
  - Set up the Python project with uv, integrated the team's branches and published the V1.0 release.
- **Gray-GOOSE — Explainable ML Evaluation under IEC 61850** *08/2025 - 04/2026*
  Tools: Python, scikit-learn, SHAP, Docker, Jupyter [ ]
  - Assembled a reproducible pipeline that trains and evaluates machine learning models against grayhole DoS attacks on IEC 61850 GOOSE traffic.
  - Generated SHAP explanations comparing runs with and without delta features, reporting classification metrics and confusion matrices for each configuration.
  - Packaged the whole experiment in Docker so other researchers reproduce the published results with a single command.

- **The Science Behind the Hit — Music Analytics Dashboard** *06/2026 - 07/2026*
  Tools: Python, Pandas, Streamlit, Docker Compose, Jupyter [GitHub]
  - Designed a bronze/silver/gold data pipeline over 2,000 Spotify tracks (2000–2019), delivering a curated analytical layer of 1,714 tracks.
  - Answered four analytical questions on what characterizes a hit (duration, explicit content, danceability and instrumentalness) on a strongly imbalanced dataset.
  - Communicated the findings in a multi-page Streamlit dashboard, containerized for one-command execution.
- **Project Osíris — Agribusiness Products and Services Platform** *09/2024 - 07/2025*
  Tools: Node.js, Express.js, SvelteKit (private repository)
  - Developed key features of the negotiation flow between platform users.
  - Took part in defining the system requirements together with the team.

Skills

- **Programming Languages:** Python, JavaScript, TypeScript, Java
- **Web & Frameworks:** React, NestJS, SvelteKit, Node.js, Express, FastAPI, HTML5, CSS3, Tailwind CSS, Bootstrap, Skeleton UI, Vite
- **Databases:** PostgreSQL, MySQL, MongoDB, ChromaDB, Firebase
- **AI & Data:** RAG, LLM agents (Agno, Groq, Gemini, Llama/Ollama), SHAP, scikit-learn, Pandas, Streamlit, Jupyter
- **Cloud & DevOps:** AWS, Azure, Docker, Docker Compose, Vercel, Render
- **Version Control & Testing:** Git, GitHub, GitLab, Git Flow, Pytest, Jest, Junit, Postman
- **Research Interests:** Cybersecurity, Explainable Artificial Intelligence, Internet of Things

Certifications

- **Web Development Course** — Udemy *12/2022*
- **Machine Learning and Data Science Course** — Udemy *12/2023*
- **Levelling Certificate, Residência em TIC Program** — BRISA *01/2025*

Additional Information

Languages: Portuguese (native), English (intermediate)